

The empirical recurrence for OEIS A235677: recovering a conjecture that is not written down

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Abstract

OEIS A235677 records “Empirical recurrence of order 87”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 127$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 87, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 87 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A235677 is “Number of $(n+1) \times (6+1)$ 0..1 arrays with the difference between each 2×2 sub-block maximum and minimum lexicographically nondecreasing columnwise and nonincreasing rowwise.”. It has offset 1 and begins

9418, 650928, 44196144, 2985166430, 201368802704, 13578805464958, 915571832243030, 61732408470815696, ...

The entry states:

Empirical recurrence of order 87 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 08:44:09, revision 7) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 641$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 127$ of the 641, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 127$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 1282$ exact terms of the model returns order 87 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{87} c_i a(n-i),$$

c_1	83	c_{23}	−170447589240294828384	c_{45}	−74999840261655371937522159	c_{67}
c_2	−565	c_{24}	1040086319641546939816	c_{46}	−83118666714311551126620200	c_{68}
c_3	−39746	c_{25}	1186129970785724035537	c_{47}	95400073682165535555409045	c_{69}
c_4	369716	c_{26}	−7793092018526094817610	c_{48}	71650700728079257354498435	c_{70}
c_5	8420525	c_{27}	−6285623727750678525275	c_{49}	−99562176514712203844257318	c_{71}
c_6	−79480823	c_{28}	47303593952581805756813	c_{50}	−49382992952729998509098051	c_{72}
c_7	−1040530434	c_{29}	24529316134777742945219	c_{51}	85700588132834500895861013	c_{73}
c_8	9302517608	c_{30}	−234365193929764081775910	c_{52}	26353080691344962573980601	c_{74}
c_9	83344898031	c_{31}	−62880228060101492956029	c_{53}	−61044175587574791651231167	c_{75}
c_{10}	−686925612372	c_{32}	953217414943934632359220	c_{54}	−10057854940868143090164756	c_{76}
c_{11}	−4565325237032	c_{33}	46719872135490203283644	c_{55}	36053252407860687785815343	c_{77}
c_{12}	34424028893261	c_{34}	−3195845743961672671017968	c_{56}	1980747328960236013957477	c_{78}
c_{13}	176752622119992	c_{35}	464616412204908079261035	c_{57}	−17674618974344632443308385	c_{79}
c_{14}	−1224942497078891	c_{36}	8856309244152867992821210	c_{58}	533715752814884181369647	c_{80}
c_{15}	−4954860272825479	c_{37}	−2839470385222573960993251	c_{59}	7194485629395552855329798	c_{81}
c_{16}	31966332318329310	c_{38}	−20314181426906645752687226	c_{60}	−690544782416821334770061	c_{82}
c_{17}	102559979963969725	c_{39}	9800572073325547169502716	c_{61}	−2430705462181950390096269	c_{83}
c_{18}	−627417888952028899	c_{40}	38571792857525846014976106	c_{62}	354451806947068222370089	c_{84}
c_{19}	−1593327267431084189	c_{41}	−24575179182611529156565452	c_{63}	680886664579153216798641	c_{85}
c_{20}	9456677571102753804	c_{42}	−60548703039150912110172401	c_{64}	−123753005647918737890974	c_{86}
c_{21}	18819809612690672392	c_{43}	47960412763763686339729600	c_{65}	−157828629822234270134676	c_{87}
c_{22}	−111379647699731485957	c_{44}	78345717487424068815345565	c_{66}	32325970695352868056197	

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 87, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $87 + 4$ terms removed returns order 87 as well, so $\deg q_{\min} = 87 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 12 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once

more on the sequence with its first $87 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 87 that this sequence satisfies — and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A235677.
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